KARTHIK S

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PROFESSIONAL SUMMARY

B.Tech graduate in Artificial Intelligence & Data Science (March 2026) with hands-on experience building end-to-end machine learning pipelines, AI-powered applications, and full-stack web solutions. Proficient in Python, XGBoost, FastAPI, Flask, and React.js. Demonstrated ability to deliver production-quality ML models achieving up to 94% accuracy and deploy intelligent systems with real-world impact. Currently expanding expertise in Generative AI, LangChain, RAG pipelines, and LLM-powered applications. Seeking AI/ML Engineer, Generative AI Engineer, or Software Developer roles where I can drive data-driven decision-making and scalable AI solutions.

EDUCATION

B.Tech in Artificial Intelligence & Data Science — 2022 – 03/2026
Sri Krishna College of Technology — Coimbatore, Tamil Nadu
CGPA: 7.02 / 10

Santiniketan Matric Hr. Sec. School — 2022
HSC - 88.67%

Santiniketan Matric Hr. Sec. School — 2020
SSLC - 96.4%

TECHNICAL SKILLS

Languages
Python, SQL, Java (Basic), HTML, CSS, JavaScript

Generative AI & LLMs
LangChain, LlamaIndex, RAG (Retrieval-Augmented Generation), LLM APIs, Pinecone, ChromaDB (Vector Databases)

Web & API Dev
FastAPI, Flask, React.js, REST APIs, Streamlit, Modular Architecture, OpenAI

Soft Skills
Problem-Solving, Cross-functional Teamwork, Agile Communication, Project Management

AI / ML Frameworks
Scikit-learn, PyTorch, XGBoost, Random Forest, Linear Regression, HuggingFace Transformers, Transformer Architecture

Data Science
Pandas, NumPy, Matplotlib, Seaborn, Feature Engineering, Hyperparameter Tuning, Model Evaluation (F1, ROC-AUC, Precision/Recall), MLflow

DevOps & Tools
PostgreSQL, Docker (Basic), Git, GitHub, Jupyter Notebook, VS Code, Google Colab, AntiGravity

PROJECTS

AI Interview Simulation System — 2026 – Present
Tech Stack: Python, LangChain, LlamaIndex, RAG, HuggingFace Transformers, FastAPI, ChromaDB, Streamlit, Docker, MLflow

- Architecting an end-to-end AI-powered interview simulation platform using LangChain and LlamaIndex to deliver dynamic, context-aware interview questions tailored to a candidate's resume and target role.
- Implementing a RAG pipeline backed by ChromaDB vector database to retrieve domain-specific knowledge and generate accurate, role-relevant follow-up questions in real time.
- Integrating HuggingFace Transformer models for candidate response evaluation, scoring communication quality, technical depth, and keyword alignment.
- Building a FastAPI backend with a Streamlit-based interactive UI, containerized using Docker for consistent cross-environment deployment.
- Tracking model experiments and evaluation metrics using MLflow to ensure reproducibility and iterative performance improvement.

Real-Time Legal Advisory System via Generative AI 12/2025 – 02/2026

Tech Stack: Python, Flask, LLM APIs, NLP, Multilingual NLP, REST API

- Architected a full-stack AI-powered legal advisory platform using Flask and LLM APIs, enabling real-time, context-aware legal guidance for non-specialist users.
- Engineered a modular microservice pipeline covering chat processing, document ingestion, and AI response generation — reducing end-to-end response latency by structuring async service calls.
- Implemented multilingual response generation and legal precedent matching, expanding platform accessibility across 3+ languages.
- Integrated document upload and parsing (PDFs/text) with voice output, enabling multimodal interaction and broadening use-case coverage.

Insurance Claims Anomaly Detection 08/2025 – 09/2025

Tech Stack: Python, XGBoost, Scikit-learn, Pandas, ROC-AUC, Feature Engineering

- Trained an XGBoost classification model to detect fraudulent insurance claims, achieving 94% accuracy on held-out test data.
- Applied advanced feature engineering and class-imbalance handling to improve minority-class detection and reduce false negatives.
- Evaluated model robustness using Precision, Recall, F1-Score, and ROC-AUC; generated performance reports for stakeholder communication.
- Reduced simulated manual claim review workload by ~30% through automated anomaly flagging in the claims pipeline.

Employee Salary Prediction System 06/2025 – 07/2025

Tech Stack: Python, Scikit-learn, Linear Regression, Random Forest, GridSearchCV

- Built a regression pipeline using Linear Regression and Random Forest to predict employee compensation, incorporating 10+ engineered features from HR datasets.
- Improved prediction accuracy by ~12% through GridSearchCV hyperparameter tuning and StandardScaler feature scaling.
- Performed comprehensive data preprocessing — null imputation, outlier detection, and categorical encoding — ensuring end-to-end data quality.

Supermarket E-Commerce Web Application 07/2024 – 08/2024

Tech Stack: React.js, JavaScript, HTML, CSS, Component Architecture

- Developed a responsive grocery e-commerce app using React.js with reusable component architecture, enabling seamless product browsing and cart management.
- Implemented advanced search, category filtering, and dynamic product rendering using React hooks for smooth, consistent UX across all flows.

RELEVANT COURSEWORK

Machine Learning & Data Science
Supervised/unsupervised learning, model evaluation, and deployment pipelines

Deep Learning Fundamentals
Neural networks, CNNs, Transformer architecture, and optimization techniques

Data Structures & Algorithms
Problem-solving with Python.

Database Management Systems
SQL, relational modeling, and query optimization

LANGUAGES

English
Professional

Tamil
Native

Telugu
Conversational